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UNIVERSITY OF UYO
DEPARTMENT OF MATHEMATICS
2025/2026 FIRST SEMESTER EXAMINATION

COURSE CODE: MTH212:**COURSE TITLE:** Introductory Algebra
TIME ALLOWED: TWO HOURS (2HRS)

Instructions: Attempt any **Three (3)** questions only.

1. (a) Define (i) the naive notion of set; (ii) the rigorous concept of set; and (iii) comment on their comparisons.
(b) (i) union of two sets A and B; (ii) intersection of two sets A and B; (iii) difference of two sets A and B; (iii) symmetric difference of two sets A and B; (iv) Equality of two sets A and B.
(c) Write short notes on (i) membership notion in set theory; (ii) concept of set inclusion; (ii) comparisons of membership and set inclusion using the property: that for any pair of objects p and A, the statements $p \in A$ and $p \in p$ cannot hold; Comment on the properties that $p \in A$ and $p \subseteq A$ cannot hold simultaneously for any pair of objects p and A. **clue:** explain the primitiveness of membership notion.

2. (a) (i) State the De Morgan's laws of complementaions and (b) prove that $(A \backslash B) \cup (B \backslash A) = (A \cup B) \backslash (A \cap B)$. (b) Prove any of the Morgans law that you know.
(c) (i) Prove that $A \backslash B = \emptyset$ if and only if $A \subseteq B$ and (ii) Prove that $A \backslash B = A$ if and only if $A \cap B = \emptyset$.

3. (a) Define the following terms:
(i) Cartesian product of two sets A and B; (ii) Binary relation on a set X; (iii) Binary operation on a set X; and (iv) the concept of mapping between two sets A and B.
(b) Define the following with examples (i) surjective mappings, (ii) injective mappings; and (iii) bijective mappings
(c) Differentiate between equivalence relations and partial odering. **Clue** Use definitions and examples.

4. (a) State the five Peano postulates. (b) Prove that the bianary relation "x is equivalent to y módulo n" denoted by $x \equiv y(\mod n)$ is an equivalence relation on the integers $\mathbb{Z}$ where $x \equiv y(modn)$ is defined if and only if $n|(x-y)$.

5. (a) Define (i) a semigroup, (ii) a monoid (iii) a group and (iv) either groupoid or abelian group.
(b) Use the table below to define and discuss the following terms: (i) commutativity; (ii) associativity; (iii) identity element; (iv) inverse element; (v) a group; (vi) abelian group

$*$	1	-1	i	$-i$
1	1	-1	i	$-i$
-1	-1	1	$-i$	i
i	i	$-i$	-1	1
$-i$	$-i$	i	1	-1